



I2C – HTU21 테스트

[CRZ Technology](#)





Mango-M32F2 I2C-HTU21 Humidity/Temperature Sensor 테스트

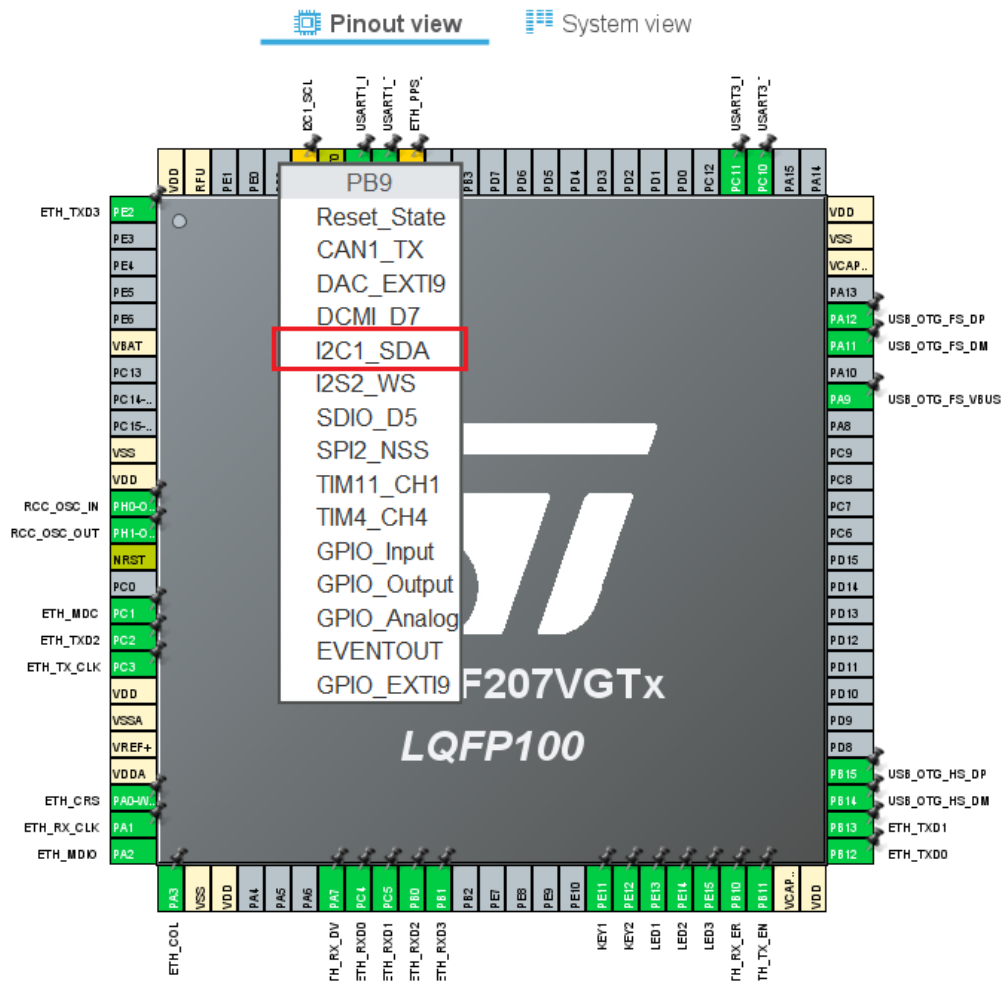
I2C pinmap

I2C – PB8, PB9

PB7	93	PB7	UART1_TX
PB8	95	PB8	I2C1_SCL
PB9	96	PB9	I2C1_SDA
PB10	47	PB10	MII_RX_FR

Pinout view

■ PB8-I2C1_SCL, PB9-I2C1_SDA 선택.



I2C 설정

- “Connectivity” 클릭, “I2C1” 클릭.
- “Mode” - “I2C” 클릭.

The screenshot displays the 'bma280_prj.ioc - Pinout & Configuration' window. The 'Pinout & Configuration' tab is active, showing a list of components on the left and configuration options on the right. In the left sidebar, under the 'Connectivity' category, 'I2C1' is selected and highlighted with a red box. The main panel shows the 'I2C1 Mode and Configuration' settings. The 'Mode' dropdown menu is set to 'I2C' and is also highlighted with a red box. Below the mode selection, there are tabs for 'Configuration', 'NVIC Settings', 'DMA Settings', 'GPIO Settings', 'Parameter Settings', and 'User Constants'. The 'Configuration' tab is active, showing a 'Reset Configuration' button and a section for 'Configure the below parameters :'. This section includes 'Master Features' with 'I2C Speed Mode' set to 'Standard Mode' and 'I2C Clock Speed (Hz)' set to '100000'. It also includes 'Slave Features' with 'Clock No Stretch Mode' set to 'Disabled' and 'Primary Address Length' set to '7 bit'.

HTU21 Test Project

■ Code 생성 후 main.c 수정

```
36  /* Private define -----*/
37  /* USER CODE BEGIN PD */
38
39  #define HTU21_I2C_ADDR (0x40 << 1)
40
41  #define REG_RTEMP_READ_HOLD          0xE3
42  #define REG_RHUM_READ_HOLD          0xE5
43  #define REG_SOFT_RESET              0xFE
44
45  //CRC error macros
46  #define CRC_MATCH                    1
47  #define CRC_NOTMATCH                0
48
49  //CRC macros
50  #define CRC_POLYNOMIAL              0x131    // Generator Polynomial  $X^8 + X^5 + X^4 + 1$ 
51  #define CRC_POLYNOMIAL_WIDTH        9
52
53  /* USER CODE END PD */
54
```

HTU21 Test Project

```
115 static int crc_check(uint16_t dt, uint8_t crc)
116 {
117     uint16_t polynomial = CRC_POLYNOMIAL << (16 - CRC_POLYNOMIAL_WIDTH);
118     uint16_t result = dt;
119     uint16_t top_bit = 0x8000;
120
121     for(int i=0;i<16;i++)
122     {
123         if(result & top_bit) result ^= polynomial;
124         result <<= 1;
125     }
126
127     result >>= 8;
128
129     return crc == result ? CRC_MATCH : CRC_NOTMATCH;
130 }
131 /* USER CODE END 0 */
```

HTU21 Test Project

```
178 reg = REG_SOFT_RESET;
179 status = HAL_I2C_Master_Transmit(&hi2c1, HTU21_I2C_ADDR, &reg, 1, 1000);
180 if(status != HAL_OK)
181 {
182     printf("status = %d\r\n", status);
183 }
184 HAL_Delay(300);
185 /* USER CODE END 2 */
186
187 /* Infinite loop */
188 /* USER CODE BEGIN WHILE */
189 while (1)
190 {
191     MX_LWIP_Process();
192
193     if(count++ % 10000 == 0)
194     {
195         reg = REG_RTEMP_READ_HOLD;
196         status = HAL_I2C_Master_Transmit(&hi2c1, HTU21_I2C_ADDR, &reg, 1, 1000);
197         if(status != HAL_OK)
198         {
199             printf("status = %d\r\n", status);
200         }
201         status = HAL_I2C_Master_Receive(&hi2c1, HTU21_I2C_ADDR, data, 3, 1000);
202         if(status != HAL_OK)
203         {
204             printf("status = %d\r\n", status);
205         }
206     }
207 }
```


HTU21 Test Project

```
206     data_raw = (data[0] << 8) | data[1];
207     crc = data[2];
208     printf("TEMP:%02x %02x %02x\r\n", data[0], data[1], data[2]);
209     if(crc_check(data_raw, crc) == CRC_MATCH)
210     {
211         data_raw &= 0xFFFC;
212         raw_value = ((17572 * data_raw) >> 16) - 4685;
213
214         printf("TEMP=%d.%dC\r\n", raw_value / 100, raw_value % 100);
215     }
216
217     reg = REG_RHUM_READ_HOLD;
218     status = HAL_I2C_Master_Transmit(&hi2c1, HTU21_I2C_ADDR, &reg, 1, 1000);
219     if(status != HAL_OK)
220     {
221         printf("status = %d\r\n", status);
222     }
223     status = HAL_I2C_Master_Receive(&hi2c1, HTU21_I2C_ADDR, data, 3, 1000);
224     if(status != HAL_OK)
225     {
226         printf("status = %d\r\n", status);
227     }
228     data_raw = (data[0] << 8) | data[1];
229     crc = data[2];
230     printf("HUMID:%02x %02x %02x\r\n", data[0], data[1], data[2]);
231     if(crc_check(data_raw, crc) == CRC_MATCH)
232     {
233         data_raw &= 0xFFFC;
234         raw_value = ((125 * data_raw) >> 16) - 6;
235
236         printf("HUMID=%d%%\r\n", raw_value);
237     }
238 }
239 /* USER CODE END WHILE */
```

Test 결과 확인

- CR-HTU21 센서 보드를 8Pin Cable을 이용 CON7에 연결함.
- 온도와 습도가 정상적으로 읽힘.

VT COM14 - Tera Term VT

File Edit Setup Control Window Help

```
I2C-HTU21 test
TEMP:66 f4 49
TEMP=23.81C
HUMID:50 22 03
HUMID=33%
TEMP:66 f4 49
TEMP=23.81C
HUMID:50 06 41
HUMID=33%
TEMP:67 00 fb
TEMP=23.84C
HUMID:50 06 41
HUMID=33%
```

Thank You

